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Original Research

Using Cattle Grazing to Restore a Rough Fescue Prairie Invaded by Kentucky Bluegrass[☆]Rafael Otfinowski^{a,*}, Hilary G. Pinchbeck^b, Peter A. Sinkins^c^a Assistant Professor, Department of Biology, University of Winnipeg, Winnipeg, MB, Canada, R3B 2E9^b Undergraduate Student, Department of Biology, University of Winnipeg, Winnipeg, MB, Canada, R3B 2E9^c Ecologist Team Leader, Parks Canada Agency, Inuvik, NT, Canada, X0E 0T0

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ABSTRACT

Rough fescue prairies were once common across the northern prairies but have now been almost entirely lost to a combination of agricultural expansion, energy development, fire suppression, and invasion by exotic species. Despite these pressures, remnant grasslands remain important in conserving biodiversity and as habitats for threatened species. In this project, we test the hypothesis that the density of Kentucky bluegrass (*Poa pratensis*), an invasive, exotic species, introduced across the Great Plains, is inversely related with the density of plains rough fescue (*Festuca hallii*), the dominant native grass of northern fescue communities, and examine whether cattle grazing could help restore invaded prairies. We tested the relationship between the two species using 33 transects (0–44 m), located within patches of rough fescue prairie at the Batoche National Historic Site of Canada in Saskatchewan. Along each transect, we recorded the density of Kentucky bluegrass and plains rough fescue and used cattle enclosures to test the hypothesis that grazing reduces the abundance of the invader and increases the abundance of plains rough fescue. Although higher densities of Kentucky bluegrass were negatively correlated with the density of plains rough fescue, grazing by cattle did not significantly reduce the density of Kentucky bluegrass, 6 yr after the initiation of grazing. However, cattle grazing also did not reduce the density of plains rough fescue, suggesting that it may provide a valuable tool to actively manage and restore invaded prairies. Our results also suggest that long-term monitoring and additional measurements of community diversity and productivity may be necessary to demonstrate the success of this restoration method.

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Introduction

Rough fescue prairies, once a dominant vegetation community in the Aspen Parkland ecoregion of western Canada, have become increasingly rare since the settlement of the prairies by European immigrants and migrants from the United States and Canada, at the end of the 19th century (Acton et al., 1998). Extirpation of plains bison (*Bison bison*), increased agricultural production, and energy development have resulted in the fragmentation and loss of rough fescue prairies and introduction of invasive species (Trottier, 1986; Campbell et al., 1994; Downing and Pettapiece, 2006). Although only 5% of the 255 000 km² of historic rough fescue prairies persist (Trottier, 2002; Holcroft-Weerstra, 2003), remnant prairies remain distinct in their botanical composition (Thorpe et al., 2015) and continue to provide important habitats for wildlife, including threatened and endangered

grassland birds (Owens and Myers, 1973; Pylypec, 1991; Godwin and Thorpe, 2002).

Suppression of historic fire cycles in the Aspen Parkland has also resulted in the widespread encroachment of trembling aspen (*Populus tremuloides* Michaux) (Weir et al., 2000), reducing the productivity of grasses (Bailey and Wroe, 1974) and the diversity of grassland communities (Moss, 1932). Remnants of fescue prairie are further threatened by a changing climate. For example, in the Cypress Hills plateau of southern Alberta and Saskatchewan, warmer spring temperatures and higher fall precipitation are increasing the establishment of lodgepole pine (*Pinus contorta* var. *latifolia* Engelmann) (Widenmaier and Strong, 2010).

Among invasive, exotic plants that threaten the integrity of fescue prairie remnants, Kentucky bluegrass (*Poa pratensis* Linnaeus) poses a high risk to their diversity (Sather, 1996; Otfinowski et al., 2007; DeKeyser et al., 2015). Introduced from Europe through seed mixtures, hay, and bedding (DeKeyser et al., 2015), Kentucky bluegrass has been widely planted for lawns, pastures, and to stabilize soil (Uchytel, 1993). It is also an important species in cultivated, perennial pastures in western Manitoba and central Saskatchewan and Alberta (Alberta Agriculture Food and Rural Development, 1981). However, once established, Kentucky bluegrass can proliferate into adjacent native

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prairies (DeKeyser et al., 2009; Thorne and Cardina, 2011; Sinkins and Otfinowski, 2012) by dispersing its prolific seeds or spreading vegetatively through rhizomes (Uchytel, 1993). Impacts of Kentucky bluegrass on native prairie communities include the accumulation of continuous mats of litter, which reduce soil temperatures and light to lower the germination of other plants, particularly those with small seeds (Bosy and Reader, 1995). Accumulation of a litter mat also creates a physical barrier that reduces the emergence of native seedlings (Williams and Crone, 2006) and may chemically suppress their germination (Bosy and Reader, 1995). Mats of roots, also created by the invading plants, further prevent water infiltration and increase water stress of native grasses (Dong et al., 2014). Finally, invasions of Kentucky bluegrass reduce the cover (White et al., 2012), richness, and diversity of invaded communities (MacDougall and Turkington, 2005; DeKeyser et al., 2015) and lead to the decline of populations of native plants, including the prairie crocus (*Anemone patens* Linnaeus), prairie junegrass (*Koeleria macrantha* [Ledebour] Schultes), thread-leaved sedge (*Carex filifolia* Nuttall), and sun sedge (*Carex inops* subsp. *heliophila* [Mackenzie] Crins) (Williams and Crone, 2006; DeKeyser et al., 2015).

Restoring fescue prairies invaded by Kentucky bluegrass is complicated by its persistence in grazed areas (Sather, 1996; Biondini et al., 1998) and its proliferation following prolonged periods of rest (Murphy and Grant, 2005; DeKeyser et al., 2009). As a result, the goal of restoration may be to reduce the vigor and spread of Kentucky bluegrass, rather than its exclusion (Godwin and Thorpe, 2004). Furthermore, the intensity of grazing must consider its impacts on the integrity of the prairie community. For example, heavy grazing of rough fescue prairies reduces their diversity and can lead to the establishment of other invasive, exotic species, including dandelion (*Taraxacum officinale* F. H. Wiggers) (Willms et al., 1985; Sinkins and Otfinowski, 2012; Thorpe, 2014). In contrast, exclusion of grazing from rough fescue prairies can also increase the proliferation of exotic invaders, including smooth brome (*Bromus inermis* Leysser) (Sinkins and Otfinowski, 2012).

In this project, we test the relationship between the density of Kentucky bluegrass and plains rough fescue (*Festuca hallii* [Vasey] Piper), the dominant native grass of rough fescue communities (Looman, 1969), and test the hypothesis that they are inversely related. Secondly, we examine whether cattle grazing could help restore invaded prairies and test the hypothesis that grazing reduces the abundance of Kentucky bluegrass and increases the abundance of plains rough fescue.

Methods

Batoche National Historic Site of Canada (Batoche NHSC) is located in central Saskatchewan (lat 52.754344, long -106.106167). In addition to its mandate to protect and present historic and cultural resources, Batoche NHSC includes areas of Great Plains rough fescue prairie (Thorpe et al., 2015), recognized as an element of special significance for the historic site (Parks Canada, 2000).

Located within the Aspen Parkland Ecoregion (Acton et al., 1998), Batoche NHSC is covered by a mix of deciduous forests and pockets of rough fescue prairies that occupy nonforested, upland sites. In North America, Great Plains rough fescue prairies occur south of the limit of tree growth in the west-central prairies and form patches of grassland within forest/woodland groves (Thorpe et al., 2015). Cooler and moister conditions promote the establishment of trembling aspen (*Populus tremuloides* Michaux) (Moss, 1932) that consolidate to form isolated groves and closed forest in the absence of fire and grazing (Anderson and Bailey, 1980; Campbell et al., 1994). Rough fescue prairies occur in the continental temperate climate with mean annual temperatures approximately 2.5°C, and extreme minimum temperatures below -40°C. Growing degree days above 5°C vary between about 1300 and 1600, and mean annual precipitation is between 350 and 500 mm (Thorpe et al., 2015). During the study period (2006–2009),

precipitation at Batoche NHSC was 393.7 ± 75.0 mm (mean $\pm$ standard deviation [SD]) (1905–2005: 388.9 ± 65.2 mm [mean $\pm$ SD]) (University of East Anglia Climatic Research Unit and Harris, 2013).

Within the west area of Batoche NHSC (689 ha), we randomly selected 23 patches of rough fescue prairie from those classified as “native grassland” in an existing vegetation map of the property (Godwin and Thorpe, 2004). Within the 23 patches of prairie, we randomly placed 17 transects (0–44 m) in the north and 17 transects in the south fields that divide the property (Godwin and Thorpe, 2004) (Fig. 1). Each transect was a minimum of 10 m from forest edges and as far as possible from trees or suckers of trembling aspen. The start and end of each transect were marked using a Global Positioning System ($\pm 2–3$ m accuracy). We recorded the frequency of Kentucky bluegrass and plains rough fescue plants 15 times along each transect (3-m intervals) using a 75×75 cm quadrat (0.563 m^2). Each quadrat was divided into 25 grid cells (5×5) and used to record the number of cells that contained one or more plants of Kentucky bluegrass or plains rough fescue. The quadrat was then flipped, end-over-end, and the counts were repeated three more times (Vogel and Masters, 2001). For each species, frequencies inside all four quadrats (2.25 m^2) were averaged per transect and scaled to 1 m^2 . This method provided a conservative estimate of plant density (Vogel and Masters, 2001). Densities of Kentucky bluegrass and plains rough fescue were measured in July, before grazing, and repeated in 2006, 2007, and 2009. Plant nomenclature follows VASCAN (Brouillet et al., 2010). Before 1990, seasonal grazing by cattle was permitted across the west area of Batoche NHSC, then stopped between 1990 and 2004, and resumed in 2004 (Godwin and Thorpe, 2004).

To test the hypothesis that grazing by cattle reduces the abundance of Kentucky bluegrass and increases the abundance of plains rough fescue, six additional transects were excluded from cattle grazing using fences (see Fig. 1). Three exclosures were constructed each in the north and south fields of Batoche NHSC in 2004. Grazing intensities and schedules were based on 44 cow/calf pairs that were moved between the north ($44 \text{ cow-calf pairs} \cdot 35 \text{ days} \cdot 40 \text{ ha}^{-1} = 1.31 \text{ AUM} \cdot \text{ha}^{-1}$) and the south fields ($44 \text{ cow-calf pairs} \cdot 24 \text{ days} \cdot 46 \text{ ha}^{-1} = 0.76 \text{ AUM} \cdot \text{ha}^{-1}$) in August or September, alternating between years (Godwin and Thorpe, 2004). All animals were moved out of the site in October.

Statistical Analyses

We used general additive models (GAMs) to test the relationship between the density of Kentucky bluegrass and plains rough fescue and to test the hypothesis that they are inversely related. Using the mean density of each species along each transect ($n = 33$), we estimated the smoothing parameter of the relationship between the two species by minimizing the generalized cross-validation criterion (Wood, 2006). One transect could not be relocated in 2009 and was dropped from the study.

We tested the hypothesis that grazing by cattle reduces the abundance of Kentucky bluegrass and increases the abundance of plains rough fescue using cattle exclosures. Each fenced transect was paired with one nearest grazed transect, situated in close proximity (< 110 m) with similar slope position and vegetative cover (see Fig. 1). Densities of Kentucky bluegrass and plains rough fescue were not normally distributed (Shapiro-Wilk test: $W = 0.89635$, P value < 0.001 and $W = 0.79933$, P value < 0.001 , respectively), and log transformations did not normalize their distributions. As a result, we examined the effect of grazing on the density of Kentucky bluegrass and plains rough fescue using linear mixed models (Bolker et al., 2009) (package “lme4” in R version: 1.1–10) and used the restricted maximum likelihood (REML) algorithm to estimate fixed-effect parameters. Fixed-effect variables in each model accounted for variance in the density of Kentucky bluegrass and plains rough fescue in response to the year since the initiation of grazing (ann: 2006, 2007, 2009) and location (field: north [higher density of invader] vs. south [lower density of invader]). Responses of Kentucky bluegrass and plains rough fescue to grazing were analyzed separately. All statistical models were implemented using R version 3.2.3 (R Core Team, 2015).

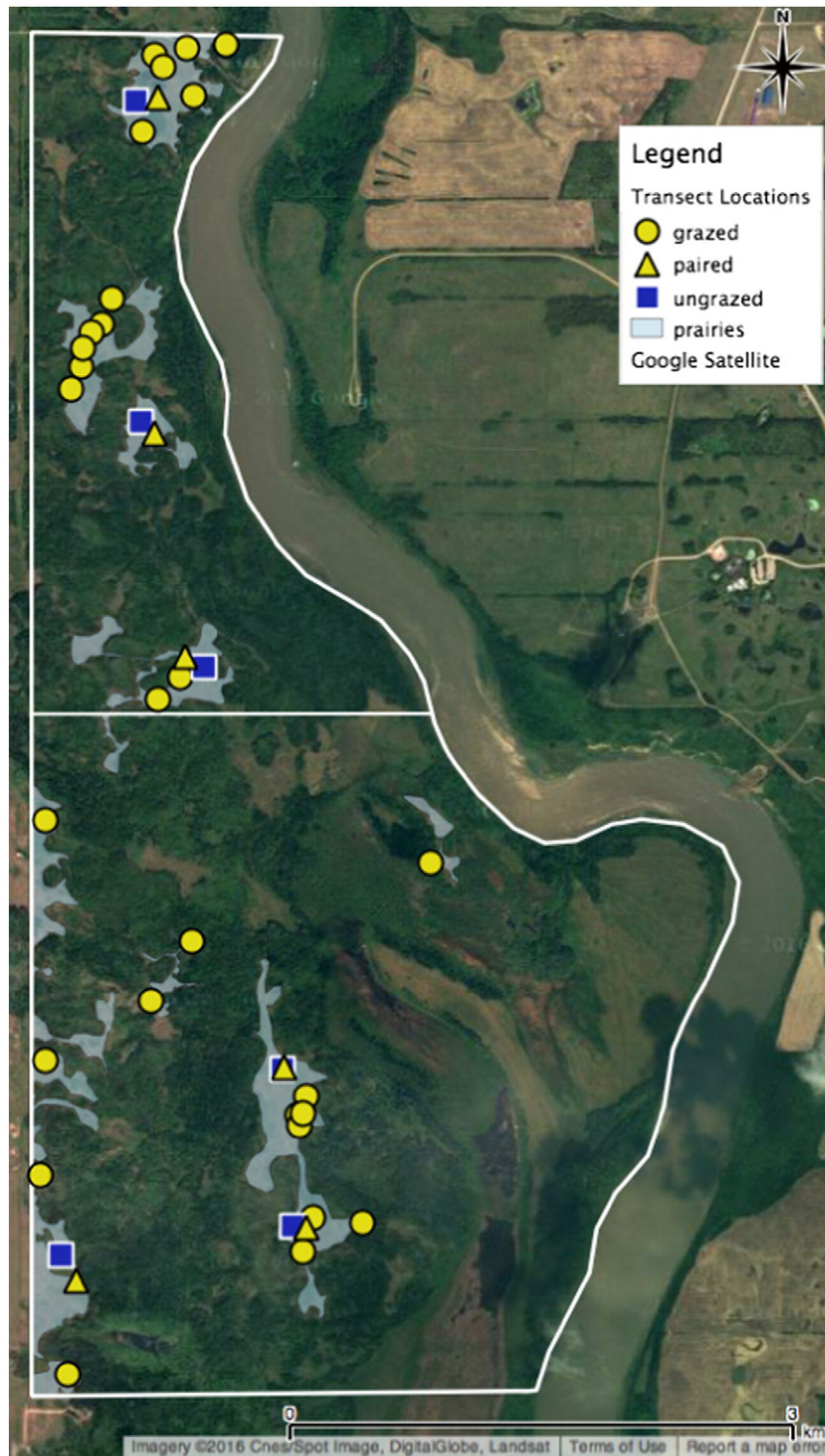


Figure 1. Grazed and ungrazed transects used to monitor the restoration of rough fescue prairies invaded by Kentucky bluegrass (*Poa pratensis* Linnaeus) at Batoche National Historic Site of Canada, Saskatchewan. Ungrazed transects (squares) were protected inside cattle enclosures and paired with grazed transects (triangles) to test the effect of grazing on restoration. Highlighted prairies were selected randomly for the experiment from an existing vegetation map (Godwin and Thorpe, 2004).

Results

Relationship Between Abundance of Kentucky Bluegrass and Plains Rough Fescue

Higher densities of Kentucky bluegrass were inversely related with the abundance of rough fescue in the north field ($R^2 = 0.71$, $P < 0.001$) (Fig. 2A). However, this impact was not significant in the south field ($R^2 = 0.08$, $P = 0.0584$) (see Fig. 2B), where the density of

Kentucky bluegrass was lower ($30.3 \text{ plants} \cdot \text{m}^{-2}$ [south field] vs. $36.1 \text{ plants} \cdot \text{m}^{-2}$ [north field]; $F_{1,115} = 19.8$, $P < 0.001$).

Effect of Grazing on Abundance of Kentucky Bluegrass and Plains Rough Fescue

Although grazing by cattle did not significantly reduce the density of Kentucky bluegrass ($p[\text{graze}] = 0.1027$; Akaike information criterion [AIC] = 83.3; Table 1), its mean density decreased between 2006

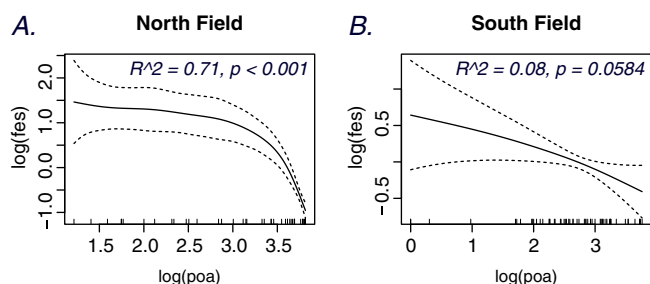


Figure 2. Density of plains rough fescue (*Festuca hallii* [Vasey] Piper) in relation to the density of Kentucky bluegrass (*Poa pratensis* Linnaeus) in the north (A) and south (B) fields of Batoche National Historic Site of Canada, Saskatchewan. General additive models were based on the density of Kentucky bluegrass and plains rough fescue, surveyed in 2006, 2007, and 2009. Dashed lines illustrate two standard errors above and below the estimates (package: mgcv [version 1.8–10], R Core Team, 2015).

(27.6 ± 7.4 plants $\cdot$ m $^{-2}$) and 2009 (15.1 ± 4.2 plants $\cdot$ m $^{-2}$) (Fig. 3A). However, the effect of grazing on Kentucky bluegrass was not significant with the year since the initiation of grazing (p[graze:ann] = 0.5926; AIC = 83.3; see Table 1). The effect of grazing on the density of Kentucky bluegrass also did not change with the location of grazing (north [higher density of invader] vs. south fields [lower density of invader]) (p[graze:field] = 0.5926; AIC = 77.5; see Table 1). Despite the lack of observed effect of cattle grazing on the density of Kentucky bluegrass, grazing also did not alter the density of plains rough fescue (p[graze] = 0.4390; AIC = 102.27; see Table 1). In contrast, its density increased between 2006 and 2009 (p[ann] < 0.001; AIC: 102.27) (see Fig. 3B) and remained higher in the north field (17.1 ± 13.1 plants $\cdot$ m $^{-2}$ vs. 3.6 ± 4.5 plants $\cdot$ m $^{-2}$) (p[field] = 0.0179; AIC = 102.73; see Table 1).

Discussion

Relationship Between Abundance of Kentucky Bluegrass and Plains Rough Fescue

Our results demonstrate an inverse relationship between the density of Kentucky bluegrass and the density of plains rough fescue. Impacts of Kentucky bluegrass on the abundance of native plants have been documented in other studies. For example, Williams and Crone (2006) observed decreased survival of prairie pasqueflower (*Anemone patens* Linnaeus) in plains rough fescue communities invaded by Kentucky bluegrass and predicted a decline in mountain rough fescue (*Festuca*

campestris Rydberg) and poverty oatgrass (*Danthonia spicata* [Linnaeus] P. Beauvois ex Roemer & Schultes) following invasion. Similarly, MacDougall and Turkington (2005) reported low cover of the native forbs, common camas (*Camassia quamash* [Pursh] Greene) and Pacific sanicle (*Sanicula crassicaulis* Poeppig ex de Candolle), in oak savanna communities dominated by Kentucky bluegrass and orchard grass (*Dactylis glomerata* Linnaeus). In mesic sand prairies, an increase in Kentucky bluegrass and smooth brome also corresponded with a decrease in the abundance of little bluestem (*Schizachyrium scoparium* [Michaux] Nash) and plains porcupine grass (*Hesperostipa spartea* [Trinius] Barkworth), as well as pale purple coneflower (*Echinacea pallida* [Nuttall] Nuttall), stiff sunflower (*Helianthus pauciflorus* Nuttall subsp. *pauciflorus*), and prairie coreopsis (*Coreopsis palmata* Nuttall) (Bowles et al., 2003). At the community level, both White et al. (2012) and DeKeyser et al. (2015) found negative correlations between the abundance of Kentucky bluegrass and the cover of native vascular plants in the foothills and Great Plains fescue prairies, respectively.

Impacts of Kentucky bluegrass on native prairie plants have been attributed to its transformation of the structure and function of invaded areas. Invading plants produce continuous mats of litter and shallow roots that decrease light and soil temperatures and reduce germination and emergence of native species (Bosy and Reader, 1995; Williams and Crone, 2006). These changes also alter surface hydrology (DeKeyser et al., 2009; Toledo et al., 2014), reduce water infiltration (Dong et al., 2014), and increase the nitrogen-to-carbon ratio of invaded soils (Hendrickson and Lund, 2010; Toledo et al., 2014; Beck et al., 2015). The scale of the impacts may be related to the continuous litter mat created by invading plants. For example, although plains rough fescue plants also produce dense litter, their cover is discontinuous, forming gaps that may allow for increased water infiltration. This homogenization of prairie environments by Kentucky bluegrass may be related to the negative correlation between its cover and plains rough fescue, observed in our study and reported in other research (e.g., White et al., 2012).

Effect of Grazing on Abundance of Kentucky Bluegrass and Plains Rough Fescue

Although our research demonstrates a decrease in the density of Kentucky bluegrass, 6 yr since the initiation of grazing, the observed decline was not significant. It is likely that the duration of our experiment was insufficient to observe a significant decline in the abundance of Kentucky bluegrass or that the low sample size increased the probability of a Type II error (Quinn and Keough, 2002). In previous studies, heavy grazing for 2 yr was necessary to restore mixed grass prairies invaded by smooth brome (Stacy et al., 2005) and Biondini et al. (1998) found that even 8 yr of heavy grazing in a mixed grass prairie dominated by Kentucky bluegrass, native species such as western wheatgrass (*Pascopyrum smithii* [Rydberg] Barkworth & D.R. Dewey), sun sedge (*Carex inops* subsp. *heliophila* [Mackenzie] Crins), and blunt sedge (*Carex obtusata* Liljeblad) had more effect on aboveground productivity than species composition. Lack of significant decline in the density of Kentucky bluegrass in our work may suggest that more intense, late-season grazing may be necessary to reduce the initial abundance of Kentucky bluegrass, followed by moderate grazing to increase the abundance of native species, including plains rough fescue (Trottier, 1986; Stacy et al., 2005; Blumenthal et al., 2012). It is also possible that the positive impact of moderate, late-season grazing could be more easily detected using other measures of community diversity (DeKeyser et al., 2015) or productivity (Biondini et al., 1998; Symstad, 2011).

The observed increase in the abundance of plains rough fescue during the study period, even in the presence of cattle grazing, suggests that grazing may be an effective tool to actively manage and restore rough fescue prairies invaded by Kentucky bluegrass. Ultimately, the impacts of grazing on invaded prairies depend on its intensity and timing (Trottier, 1986). For example, Bork et al. (2012) found that moderate, seasonal grazing of foothills rough fescue prairies, using

Table 1

Linear mixed models testing the effect of cattle grazing on the abundance of Kentucky bluegrass (*Poa pratensis* Linnaeus) and plains rough fescue (*Festuca hallii* [Vasey] Piper) in rough fescue prairies at Batoche National Historic Site of Canada, Saskatchewan. Transects ($n = 33$) were assigned as the random effect for each model. Fixed-effect variables accounted for year since the initiation of grazing (ann: 2006, 2007, 2009) and location (field: north [higher density of invader] vs. south [lower density of invader]). Linear mixed models were implemented in R version 3.2.3 (package "lme4" version 1.1–10)

Dependent variable	Fixed effects	MODEL			
		F value	DF	P	AIC
density (Poa)	graze	3.23	1,10	0.1027	83.33
	ann	2.08	2,20	0.1512	
	graze:ann	0.54	2,20	0.5926	
density (Festuca)	graze	0.65	1,10	0.4390	102.27
	ann	9.41	2,20	0.0013	
	graze:ann	0.45	2,20	0.6454	
density (Poa)	graze	2.68	1,8	0.1402	77.54
	field	0.00	1,8	0.9733	
	graze:field	0.31	1,8	0.5926	
density (Festuca)	graze	1.09	1,8	0.3264	102.73
	field	8.81	1,8	0.0179	
	graze:field	0.01	1,8	0.9167	

DF indicates model degrees of freedom; P, probability of Type I error; AIC, Akaike information criterion.

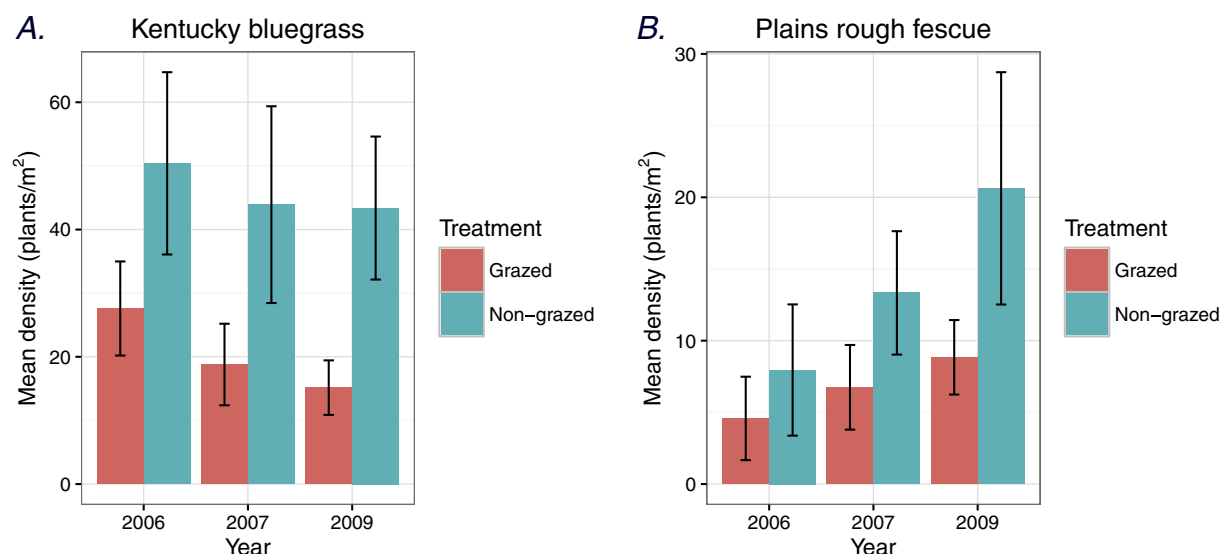


Figure 3. Mean density (plants/m² ± S.E.) of Kentucky bluegrass (*Poa pratensis* Linnaeus) (A) and plains rough fescue (*Festuca hallii* [Vasey] Piper) (B) along 12 grazed and ungrazed transects (0–44 m), recorded in 2006, 2007, and 2009 at Batoche National Historic Site of Canada, Saskatchewan. Mean densities of plants were compared using linear mixed models (package: lmer4 [version 1.1-10], R Core Team, 2015).

about 26% of the long-term forage production, reduced the abundance of foothills rough fescue (*Festuca campestris* Rydberg). Similarly, increasing grazing pressure in Great Plains fescue prairies (>1.2 animal unit months [AUM]/ha) increased the abundance of shorter and less palatable forbs and sedges (Willms et al., 1985), as well as exotic grasses, including Kentucky bluegrass, common timothy (*Phleum pratense* Linnaeus), and smooth brome (Wiloughby and Alexander 2000). This effect was especially acute during the growing season, when rough fescue was most sensitive to defoliation (Willms, 1991; Pantel et al., 2011). As a result, fall grazing, taking advantage of the late senescence of Kentucky bluegrass, may be most conducive to the conservation of rough fescue prairies (Trottier, 1986; Moisey et al., 2005). It is likely that the observed increase in the abundance of plains rough fescue in our study is attributable to the late-season grazing at Batoche NHSC. Even though recent precipitation in the northern Great Plains is higher compared with historic averages (DeKeyser et al., 2015), mean precipitation during the study period was < 1.5% higher compared with long-term data and is not likely related to the observed increase in the abundance of plains rough fescue.

Although our research focused on the abundance of Kentucky bluegrass and plains rough fescue, the use of cattle grazing to restore invaded prairies extends beyond direct defoliation of the invader. For example, Blumenthal et al. (2012) suggest that long-term grazing of northern prairie ecosystems, coevolved with heavy grazing by bison, favors short-statured, competitive native species and increases community resistance to invasion. The continuous mat of dense litter produced by Kentucky bluegrass is a major factor in reducing plant diversity, and moderate grazing, which accelerates litter decomposition (Biondini et al., 1998), could increase the establishment and proliferation of native species, including plains rough fescue. As a result, in addition to reducing the abundance of Kentucky bluegrass, grazing by cattle could be important in maintaining biotic resistance of rough fescue prairies by supporting environmental conditions suitable to the proliferation of native species (Blumenthal et al., 2012). Effects of cattle grazing on the productivity and resource capture of rough fescue communities (White et al., 2012) provide an important direction for research that could further contribute to the use of grazing to restore prairies invaded by Kentucky bluegrass.

Implications

Our results demonstrate an inverse relationship between the density of Kentucky bluegrass and plains rough fescue. In addition, the observed

increase in the abundance of plains rough fescue during the study period suggests that grazing is necessary in the maintenance or restoration of fescue prairies. However, the intensity and timing of cattle grazing remain key to the success of this tool in the restoration of rough fescue prairies. In addition, other measurements of community structure and function, including diversity and productivity, may be important to evaluate the success of grazing to restore rough fescue prairies invaded by Kentucky bluegrass. Given the relatively short duration of this study and its focus on a single site, our findings provide an opportunity to test our results at other rough fescue prairie sites in order to refine the importance of cattle grazing in the restoration of these increasingly rare ecosystems.

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